

CAPSCAN



CSE 208: Database System Lab Project

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Introduction

CapScan is a desktop software application developed to verify the authenticity and reliability of memory cards and USB drives. In today's digital world, storage devices are widely used to store critical personal, academic, and professional data. However, the market is flooded with counterfeit, low-quality, or corrupted storage devices that often misrepresent their capacity and performance. Such devices can lead to data loss, corruption, and system instability, making authenticity verification an essential step before trusting a storage device with important data.

CapScan provides a comprehensive solution by performing multiple verification processes:

1. Actual Device Capacity Verification:

Many counterfeit or low-quality storage devices falsely advertise their capacity. CapScan reads and tests the device to determine its true storage size, ensuring that users are aware of the real capacity before use. This helps prevent unexpected data loss due to insufficient storage space.

2. Data Integrity Check:

The tool performs read and write operations on the device to check for errors or data corruption. By verifying that data can be accurately stored and retrieved, CapScan ensures the integrity of user files and the reliability of the device under normal operating conditions.

3. Performance Measurement:

CapScan measures the read and write speeds of storage devices to evaluate their performance.

Low-quality components often lead to slow data transfer rates, which can affect system performance and user experience. By providing a clear performance assessment, users can make informed decisions about device usability.

Importance of Verification:

- **Protects Critical Data:** By identifying unreliable or faulty storage devices, CapScan prevents potential data loss or corruption.
- **Ensures Device Reliability:** Verification ensures that the device performs as expected under normal operating conditions.
- **Confirms True Device Capacity:** Users are provided with accurate information about the storage size, eliminating the risk of over-reported capacity.
- **Prevents Performance Issues:** Identifies slow or degraded devices caused by low-quality components, helping users avoid devices that may compromise workflow efficiency.

Overall, CapScan acts as a reliable tool for both everyday users and professionals who require a secure, trustworthy method for verifying storage devices before usage. Its comprehensive approach addresses the growing need for authenticity, reliability, and performance verification in a world increasingly reliant on digital storage solutions.

Background and Related Work

The reliability of storage devices has been a growing concern due to the widespread availability of counterfeit and low-quality memory cards, USB drives, and other storage media. Many devices in the market falsely claim higher capacities than their actual physical storage, leading to potential data loss, corruption, and system instability.

Existing verification methods often rely on specialized hardware or professional tools, which can be costly and complex for general users. Some software solutions exist, such as manufacturer-specific utilities or advanced diagnostic platforms like H2testw and CrystalDiskInfo, but they often lack a unified interface, do not provide detailed historical tracking, or require technical expertise to operate.

Recent studies emphasize the importance of combining **capacity verification, data integrity testing, and performance measurement** to ensure reliable storage device evaluation. Python-based solutions have emerged as practical alternatives due to their flexibility, cross-platform capabilities, and the availability of robust libraries for GUI design, database management, and hardware monitoring.

In this context, CapScan provides a comprehensive and user-friendly solution that integrates:

- **Capacity verification** to detect fake or over-reported devices.
- **Data integrity testing** to ensure reliable storage.

- **Performance measurement** to evaluate read/write speeds.
- **Device health monitoring** using tools such as `smartctl`.

By consolidating these features into a single desktop application with a clean interface and historical tracking, CapScan addresses gaps in existing tools, making storage device verification accessible to both casual and professional users.

Tools & Technologies

Visual Studio Code (VS Code):

Used as the primary development environment for writing, debugging, and executing the CapScan application. Its integrated features, such as syntax highlighting, version control, and debugging tools, streamline development and improve code maintainability.

Python 3:

The main programming language for CapScan, handling device detection, capacity testing, data integrity verification, GUI development, and database operations. Python's versatility and extensive libraries enable rapid and efficient development.

Tkinter:

Python's built-in GUI framework used to design a user-friendly interface. The GUI includes device selection lists, scan buttons, real-time progress indicators, and detailed result panels.

SQLite (capscan_db):

An embedded, serverless database used to store scan histories, device details, test results, and timestamps. Its lightweight design ensures portability and easy integration without requiring additional database servers.

Standard Python Modules:

- **os, platform:** For detecting connected devices and system information.
- **threading:** Enables background scanning without blocking the GUI.
- **datetime:** Generates timestamps for scan records.
- **sqlite3:** Provides access to the SQLite database for storing and retrieving scan data.

smartctl (SMART Monitoring Tools):

Command-line utility from the Smartmontools suite that:

- Reads S.M.A.R.T. data from storage devices.
- Detects device health and potential failures.
- Retrieves manufacturer-level device information.
- Improves accuracy in detecting fake or over-reported storage devices.

System Environment:

Developed and tested on **Windows 10 and Windows 11**, supporting USB drives, HDDs, SSDs, and SD cards,

ensuring broad compatibility with modern desktop configurations.

Scan History Table Schema:

Column Name	Type	Description
id	INTEGER	Unique scan record ID
device	TEXT	Device drive letter or identifier
serial	TEXT	Device serial number
claimed	REAL	Advertised capacity (MB/GB)
true	REAL	Actual verified capacity (MB/GB)
result	TEXT	Scan result (e.g., Genuine, Info)
speed	REAL	Read/Write speed (MB/s)
details	TEXT	Scan details and free space info
date	DATE TIME	Date and time of the scan

Features

- **Detection of Fake or Over-Reported Capacity:**
Identifies storage devices that falsely advertise higher capacities, ensuring that users receive genuine and reliable devices.
- **Data Integrity Verification:**
Performs read and write operations to detect corruption, safeguarding the accuracy and reliability of stored data.
- **Performance Assessment:**
Measures device read and write speeds to evaluate performance and identify slow or low-quality storage media.
- **Device Health Monitoring:**
Uses S.M.A.R.T. data to check the overall health of larger storage devices, allowing early detection of potential failures.
- **Scan History Management:**
Maintains a record of previous scans, including device details and test results, for easy tracking and reference.

Results and Limitations

Results:

- **Capacity Verification:** Sample scans accurately display the actual storage capacity of devices compared to their advertised values. This allows users to immediately identify over-reported or fake devices.
- **Detection of Corruption and Fake Devices:** The system effectively identifies devices with corrupted sectors or counterfeit specifications, preventing potential data loss.
- **Performance Measurement:** Read and write speeds are measured and reported, enabling users to assess the operational performance of each device and detect low-quality storage media.
- **Device Health Monitoring:** For larger storage devices that support S.M.A.R.T. data, the application provides detailed health reports, allowing early detection of possible failures and reducing the risk of sudden data loss.
- **Historical Records:** Scan results are saved in the embedded database, allowing users to track device performance and reliability over time.

Limitations:

- **Time-Consuming Testing:** Scanning large-capacity devices can take significant time, especially when performing read/write verification and performance

tests.

- **Hardware Dependence:** The accuracy of performance measurements and device detection can vary depending on the host system's hardware, such as USB controllers or disk interfaces.
- **System Permissions:** Proper system permissions are required to access certain devices, especially when reading S.M.A.R.T. data or performing write tests.
- **Limited Health Checks:** Health monitoring using S.M.A.R.T. data is only available for devices that support it, typically larger HDDs and SSDs. Smaller devices such as low-capacity USB drives may not provide sufficient health information.
- **Platform Restriction:** The application has been developed and tested only on Windows 10 and Windows 11, limiting compatibility with other operating systems.

Conclusion and Future Enhancements

The CapScan project successfully delivers a comprehensive solution for verifying the authenticity, capacity, and performance of storage devices, including memory cards and USB drives. By performing real vs. advertised capacity checks, data integrity tests, performance measurements, and health monitoring for supported devices, CapScan ensures data safety and reliable device operation.

The user-friendly interface, combined with scan history management, allows users to easily track previous results and make informed decisions about their storage devices. Overall, CapScan addresses the critical problem of fake or corrupted storage media in an accessible and practical manner for both general and professional users.

Future Enhancements:

To further improve functionality and user experience, the following enhancements are proposed:

- 1. Cloud-Based Report Syncing:**

Enable users to store and access scan results across multiple devices, providing remote access and centralized record management.

- 2. Mobile Application Support:**

Extend CapScan to mobile platforms, allowing users to verify storage devices and monitor health on-the-go.

- 3. Expanded Device Database:**

Include a broader range of device models to improve compatibility and accuracy in authenticity verification.

- 4. Real-Time Model Verification and Updates:**

Implement automated updates for device models and firmware recognition, ensuring accurate detection during scans.

- 5. Advanced Predictive Health Monitoring:**

Use S.M.A.R.T. data and historical trends to predict potential device failures, allowing users to take preventive action to safeguard important data.

Through these enhancements, CapScan can evolve into a more versatile and intelligent tool, providing proactive

monitoring, broader device coverage, and seamless access across platforms while maintaining its core functionality of reliable storage verification.